Inauguration of solar roads in France (Source: nbcnews.com)

water puddles and poor-quality roads would decrease.

Solar structures for roads have not been explored much even at global scale. There have been a few cases of solar road installation in the recent years that have started to deliver.

In 2014, the Netherlands chiseled a small stretch of biking road with solar panel material. It was a test drive of solar panel roads, covering only 70 metres. The setup consists of durable solar trapping material, engraved inside glass coating, with more layers of materials to create friction while driving.

Annually, the 70m stretch generates 70kWh energy per square metre of road. Energy generation in the first six months was 3000 kilowatt-hour (kWh), enough to run two small households for a year. After one complete year, the total energy production amounted to 9800kWh, which is enough to power three average-scale Dutch households annually. The project was expanded by 20m in 2016 and is still under further expansion.

While this was a comparatively smaller project, solar panel roads saw a larger implementation for the first time in France in 2016. One kilometre of a highway in Tourouvre au Perche village is covered by 2800sq.m of solar panels, protected under coats of resin, polymer and silicon. Every day, more than 2000 cars (including heavy vehicles) drive over this road. Yet, the road has been successfully able to carry this extent of load without any hiccups. It is also resistant to the heat generated regularly. Annual electricity generation is 280 megawatt-hours (MWh), which is sufficient to power streetlights and other adjoining infrastructure in the area. Notably, the village receives less-than-moderate sunlight throughout the year. That said, 280MWh electricity generation per year across 1km length is quite impressive. The French administration is planning road expansions to power a substantial fraction of the country's public properties as well as residential constructions with the help of solar energy captured by these roadways.

China has also joined the race by designing its own solar panel roads. Last year, in the Jinan region, it revealed 1km stretch of a two-way road, embedded with 5875sq.m solar surface. The road is expected to generate one million kilowatt-hours of power in a year. This energy, sufficient to power about 800 homes in China, will be used to power various infrastructures including streetlights, billboards, surveillance cameras and even toll booths in the area. An intelligent heat-capture system of the solar setup will also keep snow accumulation in check during winters.

The challenges

Solar panel road technology is in nascency with more players slowly joining the movement. All the aforementioned projects have been successful, but their execution was far from easy.

The first major challenge is the huge cost of investment. The 1km long French project cost nearly $6 million. China's project cost $458 per square metre, which totals to approximately

Solar surface on a China road (Source: www.news.cn)